Ryan Mahshie

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Education

Northeastern University *(Boston, MA)* Sep 2023 – May 2027

Bachelor of Science in Computer Science, AI Concentration ***3.91 GPA*** *(Dean's List) | Merit Scholarship*

Activities: Cybersecurity CTF Club, Peer2Peer CS Mentor, Downhillers Ski Club

Experience

Software Engineer (Intern) *(Palo Alto, California)* Sep 2026 – Dec 2026

Amazon Web Services

- Developed a Java service on EC2 with asynchronous queue pollers and AWS CDK infrastructure that connects ML models to EMR Managed Scaling, enabling Spark clusters to scale on predicted demand instead of current load
- Identified a concurrency bug in Managed Scaling's SQS message processing, where blocking task submission left 95% of each worker pool idle; findings escalated to owning team
- Rewrote the team's local test harness for EMR Managed Scaling as portable scripts, letting engineers validate scaling changes in 5 minutes before merge instead of a 1+ hour post-merge pipeline run

Software Engineer (Intern) *(Bellevue, Washington)* Jun 2026 – Aug 2026

Visa

- Built and deployed a Java/Vert.x model-serving API that combines XGBoost4J inference with merchant configuration from internal APIs to rank eligible payment methods in real time
- Engineered an asynchronous XGBoost4J training pipeline over 10M+ transactions with temporal cross-validation, job-status polling, crash recovery, and versioned model publication
- Designed a pluggable post-inference layer that applies device, fee, and dispute adjustments to rankings without model retraining

Software Engineer (Co-op) *(Alexandria, Virginia)* Jul 2025 – Dec 2025

OpsCanvas

- Architected and implemented Azure cloud support in Go/Huma with API endpoints for installation management, database integration, and automated DataSync task orchestration, doubling supported cloud platforms
- Refactored backend API logic into a shared Python SDK, reducing duplicate code by 60% and enabling consistent service integration across cloud environments
- Built GitHub Actions integration tests and managed AWS EKS deployments to validate Azure support, preventing breaking changes from reaching production

AI Software Engineer (Intern) *(McLean, Virginia)* Jun 2025 – Jul 2025

Pelagic AI

- Designed 8-node cyclic LangGraph agents with iterative tool use and state review, enabling adaptive aircraft movement decisions throughout simulations
- Developed custom MCP tools to compress and filter LLM outputs, enabling accurate spatial reasoning over large airport graphs within strict context limits

Selected Technical Projects

LawSearch AI | *Python, LangChain, LangGraph, ChromaDB*

- Built RAG system over 2,000+ pages of federal spending bills where every figure in the final answer, including synthesized aggregates, links to its source passage as a hover citation
- Designed LangGraph routing and parallel map-reduce retrieval, cutting typical latency from 3 min to 30 seconds
- Built a 25-query end-to-end RAG benchmark with source-backed gold references, LLM judging, and stage diagnostics; scored 9.1/10 with 92.4% weighted fact recall and 100% provenance pass rate

Technical Skills

Languages: Java, Python, Go, TypeScript, SQL, JavaScript
Systems & Tools: AWS (EMR, EC2, SQS, DynamoDB, CDK), Docker, Kubernetes, Postgres, MongoDB, Oracle Database, Azure, GitHub Actions
Frameworks & Libraries: Spring, Vert.x, XGBoost, FastAPI, LangChain, LangGraph, React, Pydantic, ChromaDB